

EMERALD, Australia

WMO: 943630

Lat: 23.5694S

Lon: 148.1756E

Elev: 190

StdP: 99.06

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	7.4	-5.7	2.4	18.2	-4.0	2.8	17.5	9.0	16.3	8.3	16.2	2.4	160	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	38.0	21.6	36.6	21.4	35.2	21.2	25.2	29.3	24.6	29.1	24.1	28.9	4.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.0	25.9	23.7	19.0	25.6	23.1	18.2	25.5	78.4	28.8	75.9	29.2	73.8	28.8	35.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	8.0	7.2	DB	3.2	40.6	1.2	1.7	2.3	41.9	1.6	42.9	0.9	43.9	0.0	45.2	
			WB	1.0	27.1	1.3	2.5	0.1	28.9	-0.7	30.4	-1.4	31.8	-2.4	33.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	28.1	27.5	26.4	23.3	19.5	16.6	16.1	17.6	21.1	24.3	26.3	27.8
	DBStd	4.97	2.12	2.31	2.01	2.16	2.52	2.65	2.44	2.56	2.59	2.58	2.37	2.51
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	208	0	0	0	1	16	64	77	43	6	1	0	0
	CDD10.0	4698	562	491	507	399	296	198	189	237	334	443	489	552
	CDD18.3	1863	304	258	248	151	54	12	7	22	90	185	239	293
	CDH23.3	19882	3521	2814	2536	1262	404	88	75	288	999	1925	2573	3396
(14) Wind	WSAvg	3.7	4.1	3.9	3.8	3.5	3.3	3.4	3.3	3.4	3.5	4.0	4.1	4.2
	PrecAvg	646	96	96	61	36	52	21	28	21	19	38	65	112
	PrecMax	1075	424	266	309	167	258	111	158	131	83	124	144	286
	PrecMin	344	4	10	2	0	0	0	0	0	0	0	2	22
	PrecStd	176	79	69	60	40	60	26	35	32	23	37	40	68
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	39.4	37.0	33.4	30.4	27.0	27.0	30.6	35.0	37.8	38.5	40.1
		MCWB	22.7	22.1	21.5	20.8	18.9	17.4	17.0	16.4	17.6	19.1	20.9	21.1
	2%	DB	37.2	36.6	34.7	31.7	28.3	25.5	25.2	28.0	32.3	35.0	36.2	37.8
		MCWB	22.5	22.5	21.6	19.8	17.8	16.8	15.7	16.1	17.2	19.0	20.3	21.3
	5%	DB	35.5	34.8	33.4	30.4	27.0	24.3	23.9	26.3	30.3	33.1	34.7	36.0
		MCWB	22.2	22.3	21.3	19.3	17.2	16.1	15.1	15.5	16.9	18.8	20.2	21.3
	10%	DB	34.1	33.3	32.1	29.2	25.7	22.9	22.6	24.7	28.6	31.3	33.0	34.3
		MCWB	22.1	22.1	20.9	18.9	16.8	15.4	14.5	14.8	16.5	18.4	19.8	21.0
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	26.1	25.3	23.7	21.6	20.5	19.7	19.8	22.2	23.0	24.1	25.2
		MCDB	30.2	28.5	28.7	27.7	25.8	22.9	23.5	25.2	25.5	28.5	30.8	29.3
	2%	WB	25.2	25.1	24.5	22.4	20.1	18.8	18.4	18.4	20.4	22.0	23.1	24.3
		MCDB	29.9	29.0	28.1	26.9	24.9	22.4	22.1	23.9	25.0	27.8	29.4	29.6
	5%	WB	24.6	24.5	23.8	21.6	19.1	17.7	17.0	17.3	19.2	21.1	22.3	23.7
		MCDB	29.4	28.9	27.8	26.5	24.3	22.0	21.0	23.1	25.2	27.1	28.9	29.4
	10%	WB	24.0	24.0	23.1	20.7	18.2	16.7	15.8	16.3	18.2	20.4	21.7	23.1
		MCDB	28.8	28.9	27.8	25.8	23.4	21.1	20.3	22.3	25.2	26.8	28.4	28.7
(36) Mean Daily Temperature Range	5% DB	MDBR	11.2	10.7	11.3	11.8	12.3	12.1	13.3	14.4	14.6	13.6	12.6	11.9
		MCDBR	13.5	13.0	13.0	13.0	13.2	13.3	14.4	16.2	17.0	16.1	14.9	14.3
		MCWBR	3.4	3.3	3.3	4.4	5.5	6.2	6.9	6.8	5.9	4.9	4.2	3.8
	5% WB	MCDBR	10.4	9.9	9.6	10.4	10.8	10.6	11.4	13.5	12.7	12.5	12.0	11.3
		MCWBR	3.6	3.4	3.4	4.0	4.7	5.2	5.8	5.9	5.2	4.5	4.0	3.7
(40) Clear-Sky Solar Irradiance	taub		0.386	0.372	0.352	0.336	0.316	0.306	0.297	0.311	0.345	0.373	0.389	0.390
	taud		2.508	2.552	2.597	2.605	2.600	2.638	2.637	2.576	2.482	2.438	2.435	2.468
	Ebn at Noon		958	958	951	924	904	896	917	937	942	946	949	955
	Edh at Noon		115	108	98	90	83	76	79	91	108	119	123	120
(44) All-Sky Solar Radiation	RadAvg		6.83	6.50	5.90	5.19	4.34	3.73	4.20	5.13	6.11	6.87	7.10	7.13
	RadStd		0.48	0.56	0.48	0.47	0.22	0.29	0.31	0.31	0.54	0.43	0.77	0.65

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page